

DATA EXPLORATION – ASSIGNMENT 2

DATA CLEANING

Data cleaning, also known as data cleansing or data scrubbing, is a fundamental process in data preparation. It is the process of identifying and correcting errors, inconsistencies, inaccuracies in datasets to improve data quality and integrity.

1. Handling Missing Values:

Missing data is a common issue in datasets and can affect decision making. It can lead to incorrect conclusions, thereby affecting the quality of analysis.

- **Missing values in the 'Price' column**

To find the missing values in the price column, initially I have used COUNTBLANK formula to count the number of cells which has blank value.

	A	B	C	D	E	F
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
2	28-JAN-US	Laptop	Dell	1000	30	Electronics
3	15-FEB-US	Sneakers	Nike	80	15	Fashion
4	03-MAR-US	Coffee Maker	Keurig	130	40	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics
6	22-MAY-US	Backpack	North Face		20	Outdoor
7	07-JUN-UK	Headphones	Sony	200	45	Electronics
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen
10	05-SEP-UK	Tablet	Apple	500	50	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
12	07-JUN-UK	Headphones	Sony	200	45	Electronics

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category							
2	28-JAN-US	Laptop	Dell	1000	30	Electronics							
3	15-FEB-US	Sneakers	Nike	80	15	Fashion							
4	03-MAR-US	Coffee Maker	Keurig	130	40								
5	11-APR-US	Smartphone	Samsung	900	25	Electronics							
6	22-MAY-US	Backpack	North Face		20	Outdoor							
7	07-JUN-UK	Headphones	Sony	200	45	Electronics							
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion							
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen							

L2													: X ✓ fx		=COUNTBLANK(D2:D35)											
	A	B	C	D	E	F	G	H	I	J	K	L														
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category																				
2	28-JAN-US	Laptop	Dell	1000	30	Electronics					COUNTBLANK		6													
3	15-FEB-US	Sneakers	Nike	80	15	Fashion																				
4	03-MAR-US	Coffee Maker	Keurig	130	40																					
5	11-APR-US	Smartphone	Samsung	900	25	Electronics																				
6	22-MAY-US	Backpack	North Face		20	Outdoor																				
7	07-JUN-UK	Headphones	Sony	200	45	Electronics																				
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion																				
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen																				
10	05-SEP-UK	Tablet	Apple	500	50	Electronics																				

Once I found the number of blanks, I used ISBLANK with If condition to fill the blank price their mean values as shown below. I have used absolute reference (\$) to hold the correct value. Then, dragged the autofill for entire column.

Clipboard		Font		Alignment			
G2		=IF (ISBLANK(D2), AVERAGE(\$D\$2:\$D\$35), D2)					
	A	B	C	D	E	F	G
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$).
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	
4	03-MAR-US	Coffee Maker	Keurig	130	40		
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	
6	22-MAY-US	Backpack	North Face		20	Outdoor	
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	

Result:

G2								=IF (ISBLANK(D2), AVERAGE(\$D\$2:\$D\$35), D2)	
	A	B	C	D	E	F	G	H	
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$).		
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000		
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	80		
4	03-MAR-US	Coffee Maker	Keurig	130	40		130		
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	900		
6	22-MAY-US	Backpack	North Face		20	Outdoor	298		
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	200		
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30		
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90		
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	500		
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130		

- **Products with missing categories**

In order to fill the product categories which has missing values, I have used ISBLANK with If condition to impute the blank cells with text as “unknown”.

Format Painter								
Clipboard			Font		Alignment		Number	
SUM			=IF(ISBLANK(F2),"Unknown",F2)					
	A	B	C	D	E	F	G	H
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)	Category.
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	F2)
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	
4	03-MAR-US	Coffee Maker	Keurig	130	40		130	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	900	
6	22-MAY-US	Backpack	North Face		20	Outdoor	298	
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	200	

Result:

Clipboard			Font		Alignment		Number	
H4			=IF(ISBLANK(F4),"Unknown",F4)					
	A	B	C	D	E	F	G	H
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)	Category.
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	Electronics
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	Fashion
4	03-MAR-US	Coffee Maker	Keurig	130	40		130	Unknown
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	900	Electronics
6	22-MAY-US	Backpack	North Face		20	Outdoor	298	Outdoor
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	200	Electronics
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30	Fashion
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90	Kitchen

2. Correcting Inconsistent Data:

Inconsistent data including variations in text formats, typos and misspellings can hinder analysis. Standardizing text formats is crucial to ensure accurate data in data analytics. To achieve this we are using a powerful excel tool “Find and Replace” to quickly address these issues.

Identifying the Errors:

- I could find there is inconsistent data in “Product Name” column as it has upper- and lower-case inconsistencies.
- I could find there is spelling mistakes in “Product Category” column.

Find and Replace Function:

I have used short cut key CTRL + H to use find and replace

Product Name with Inconsistency:

	A	B	C	D	E	F
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
2	28-JAN-US	Laptop	Dell	1000	30	Electronics
3	15-FEB-US	Sneakers	Nike	80	15	Fashion
4	03-MAR-US	coffee Maker	Keurig	130	40	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics
6	22-MAY-US	Backpack	North Face		20	Outdoor
7	07-JUN-UK	Headphones	Sony	200	45	Electronics
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen
10	05-SEP-UK	Tablet	Apple	500	50	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
12	07-JUN-UK	Laptop	Sony	200	45	Electronics
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion
14	08-DEC-DE	coffee Maker	Nespresso	120	35	Kitchen
15	18-FEB-CA	Smartwatch	Fitbit		15	Electronics

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category					
2	28-JAN-US	Laptop	Dell	1000	30	Electronics					
3	15-FEB-US	Sneakers	Nike	80	15	Fashion					
4	03-MAR-US	coffee Maker	Keurig	130	40						
5	11-APR-US	Smartphone	Samsung	900	25	Electronics					
6	22-MAY-US	Backpack	North Face		20	Outdoor					
7	07-JUN-UK	Headphones	Sony	200	45	Electronics					
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion					
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen					
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics			
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor			
12	07-JUN-UK	Laptop	Sony	200	45	Electronics	200	Electronics			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	90	Fashion			
14	08-DEC-DE	coffee Maker	Nespresso	120	35	Kitchen	120	Kitchen			
15	18-FEB-CA	Smartwatch	Fitbit		15	Electronics	298	Electronics			

Result: Product Name with Proper Format:

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category					
2	28-JAN-US	Laptop	Dell	1000	30	Electronics					
3	15-FEB-US	Sneakers	Nike	80	15	Fashion					
4	03-MAR-US	Coffee Maker	Keurig	130	40						
5	11-APR-US	Smartphone	Samsung	900	25	Electronics					
6	22-MAY-US	Backpack	North Face		20	Outdoor					
7	07-JUN-UK	Headphones	Sony	200	45	Electronics					
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion					
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen					
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics			
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor			
12	07-JUN-UK	Laptop	Sony	200	45	Electronics	200	Electronics			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	90	Fashion			
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	120	Kitchen			
15	18-FEB-CA	Smartwatch	Fitbit		15	Electronics	298	Electronics			

Product Category with Spelling Mistakes:

	A	B	C	D	E	F
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category
2	28-JAN-US	Laptop	Dell	1000	30	Electronics
3	15-FEB-US	Sneakers	Nike	80	15	Fashion
4	03-MAR-US	Coffee Maker	Keurig	130	40	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics
6	22-MAY-US	Backpack	North Face		20	Outdoor
7	07-JUN-UK	Headphones	Sony	200	45	Electronics
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen
10	05-SEP-UK	Tablet	Apple	500	50	Electronics
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
12	07-JUN-UK	Laptop	Sony	200	45	Electronics
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
15	18-FEB-CA	Smartwatch	Fitbit		15	Electronics
16	16-APR-ES	Headphones	Bose	250	20	Electronics
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)	Category			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	Electronics			COUNTBLANK
3	15-FEB-US	Sneakers	Nike	80	15	Fashion					
4	03-MAR-US	Coffee Maker	Keurig	130	40						
5	11-APR-US	Smartphone	Samsung	900	25	Electronics					
6	22-MAY-US	Backpack	North Face		20	Outdoor					
7	07-JUN-UK	Headphones	Sony	200	45	Electronics					
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion					
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen					
10	05-SEP-UK	Tablet	Apple	500	50	Electronics					
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor					
12	07-JUN-UK	Laptop	Sony	200	45	Electronics	200	Electronics			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	90	Fashion			
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	120	Kitchen			
15	18-FEB-CA	Smartwatch	Fitbit		15	Electronics	298	Electronics			
16	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics			
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories			
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	160	Electronics			

Find and Replace

Find

Replace

Find what: Electronics

Replace with: electronics

Options >>

Replace All Replace Find All Find Next Close

Result: Product Category without Spelling Mistakes:

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)	Category			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	Electronics			COUNTBLANK
3	15-FEB-US	Sneakers	Nike	80	15	Fashion					
4	03-MAR-US	Coffee Maker	Keurig	130	40						
5	11-APR-US	Smartphone	Samsung	900	25	Electronics					
6	22-MAY-US	Backpack	North Face		20	Outdoor					
7	07-JUN-UK	Headphones	Sony	200	45	Electronics					
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion					
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen					
10	05-SEP-UK	Tablet	Apple	500	50	Electronics					
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor					
12	07-JUN-UK	Laptop	Sony	200	45	Electronics	200	Electronics			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	90	Fashion			
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	120	Kitchen			
15	18-FEB-CA	Smartwatch	Fitbit		15	Electronics	298	Electronics			
16	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics			
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories			
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	160	Electronics			
19	07-JAN-IT	Laptop	Asus	800	40	Electronics	800	Electronics			

Microsoft Excel

All done. We made 1 replacements.

OK

3. Removing Duplicates:

Dealing with duplicates is a critical aspect for data cleaning ensuring the dataset is free of inaccuracies. It contains unnecessary storage space and removing them will optimize data storage and reduces the size of overall data sets.

Select the entire table->Go to “Data” tab->Click on Remove Duplicates->In the remove Duplicates dialogue box, select the columns to check for duplicates and click OK.

Note: I have just added a duplicate row of data's in order to show an example

8	11-JUL-CA	Blender	Vitamix	400	40	Kitchen
9	16-APR-ES	Headphones	Bose	230	20	Electronics
0	07-MAR-CA	Dress	Zara	60	30	Fashion
1	13-APR-CA	Toaster	Hamilton		10	Kitchen
2	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
3	02-DEC-CA	Jeans	Levi's		50	Fashion
4	17-JUN-IN	Laptop	HP	950	25	
5	09-JUL-FR	Watch	Casio	100	20	Accessories
6	13-APR-CA	Toaster	Hamilton		10	Kitchen
7						

	A	B	C	D	E	F	G
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	100
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	8
4	03-MAR-US	Coffee Maker	Keurig	130	40		13
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	90
6	22-MAY-US	Backpack	North Face		20	Outdoor	20
7	07-JUN-UK	Headphones					
8	19-JUL-UK	T-shirt					
9	23-AUG-UK	Blender					
10	05-SEP-UK	Tablet					
11	14-OCT-UK	Hiking Boots					
12	07-JUN-UK	Laptop					
13	25-NOV-AU	Sneakers					
14	08-DEC-DE	Coffee Maker					
15	18-FEB-CA	Smartwatch					
16	16-APR-ES	Headphones					
17	21-AUG-CA	Laptop Bag					
18	20-AUG-CN	Smartwatch					
19	27-JAN-IT	Laptop					
20	01-MAR-UK	Sunglasses	Oakley		15	Fashion	29
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	20
22	11-MAR-US	Smartwatch	Apple	300	10	Electronics	30

Remove Duplicates

To delete duplicate values, select one or more columns that contain duplicates.

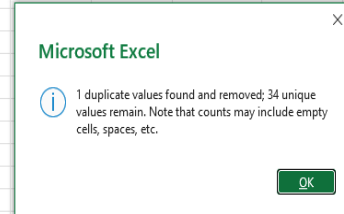
☒ My data has headers

Columns

- ☒ Product ID
- ☒ Product Name
- ☒ Brand Name
- ☒ Price (\$)
- ☒ Quantity
- ☒ Category

Result: Duplicates Removed (36th Row):

	A	B	C	D	E	F	G	H	I	J	K	L
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)	Category				
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	1000	Electronics			COUNTBLANK	6
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	80	Fashion				
4	03-MAR-US	Coffee Maker	Keurig	130	40		130	Unknown				
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	900	Electronics				
6	22-MAY-US	Backpack	North Face		20	Outdoor	298	Outdoor				
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	200	Electronics				
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	30	Fashion				
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	90	Kitchen				
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	500	Electronics				
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	130	Outdoor				
12	07-JUN-UK	Laptop	Sony	200	45	Electronics	200	Electronics				
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	90	Fashion				
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	120	Kitchen				
15	18-FEB-CA	Smartwatch	Fitbit		15	Electronics	298	Electronics				
16	16-APR-ES	Headphones	Bose	250	20	Electronics	250	Electronics				
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	50	Accessories				
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	160	Electronics				
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	980	Electronics				
20	01-MAR-UK	Sunglasses	Oakley		15	Fashion	298	Fashion				
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	200	Outdoor				
22	14-AUG-US	Camera	Coleman	200	10	Outdoor	200	Outdoor				
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	80	Kitchen				
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	150	Electronics				
25	21-AUG-CA	Laptop Bag	Samsonite		35	Accessories	298	Accessories				
26	29-SEP-CA	Smartphone	Google	800	45		800	Unknown				
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	130	Fashion				
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	400	Kitchen				
29	16-APR-ES	Headphones	Bose	230	20	Electronics	230	Electronics				
30	07-MAR-CA	Dress	Zara	60	30	Fashion	60	Fashion				
31	13-APR-CA	Toaster	Hamilton		10	Kitchen	298	Kitchen				
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	130	Electronics				
33	02-DEC-CA	Jeans	Levi's		50	Fashion	298	Fashion				
34	17-JUN-IN	Laptop	HP	950	25		950	Unknown				
35	09-JUL-FR	Watch	Casio	100	20	Accessories	100	Accessories				
36												



4. Splitting and Merging Data:

Splitting Data:

Splitting data is useful when combined data is combined in a single cell and needs to be separated for better analysis or organization.

Here I am going to split the "Product ID" column into two separate columns for "Manufacturing Date" and "Country Code".

Select the range of cells->Go to data tab and click "text to columns"->provide appropriate delimiter and format.

Splitting the "Product ID" column into two separate columns for "Manufacturing Date" and "Country Code":

	A	B	C	D	E	F	G	H
1	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Price (\$)	Quantity	Category
2	28-JAN-US			Laptop	Dell	1000	30	Electronics
3	15-FEB-US			Sneakers	Nike	80	15	Fashion
4	03-MAR-US			Coffee Maker	Keurig	130	40	
5	11-APR-US			Smartphone	Samsung	900	25	Electronics
6	22-MAY-US			Backpack	North Face		20	Outdoor
7	07-JUN-UK			Headphones	Sony	200	45	Electronics
8	19-JUL-UK			T-shirt	Adidas	30	5	Fashion
9	23-AUG-UK			Blender	Ninja	90	35	Kitchen
10	05-SEP-UK			Tablet	Apple	500	50	Electronics
11	14-OCT-UK			Hiking Boots	Timberland	130	10	Outdoor
12	07-JUN-UK			Laptop	Sony	200	45	Electronics
13	25-NOV-AU			Sneakers	Adidas	90	40	Fashion
14	08-DEC-DE			Coffee Maker	Nespresso	120	35	Kitchen
15	18-FEB-CA			Smartwatch	Fitbit		15	Electronics
16	16-APR-ES			Headphones	Bose	250	20	Electronics
17	21-AUG-CA			Laptop Bag	Samsonite	50	35	Accessories
18	20-AUG-CN			Smartwatch	Huawei	160	15	Electronics
19	27-JAN-IT			Laptop	Asus	980	10	Electronics
20	01-MAR-UK			Sunglasses	Oakley		15	Fashion

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)	Category
2	28-JAN-US	28-JAN-US			Laptop	Dell	1000	30	Electronics	1000	Electronics
3	15-FEB-US	15-FEB-US			Sneakers	Nike	80	15	Fashion	80	Fashion
4	03-MAR-US	03-MAR-US			Coffee Maker	Keurig					
5	11-APR-US	11-APR-US			Smartphone	Samsung					
6	22-MAY-US	22-MAY-US			Backpack	North Face					
7	07-JUN-UK	07-JUN-UK			Headphones	Sony					
8	19-JUL-UK	19-JUL-UK			T-shirt	Adidas					
9	23-AUG-UK	23-AUG-UK			Blender	Ninja					
10	05-SEP-UK	05-SEP-UK			Tablet	Apple					
11	14-OCT-UK	14-OCT-UK			Hiking Boots	Timberland					
12	07-JUN-UK	07-JUN-UK			Laptop	Sony					
13	25-NOV-AU	25-NOV-AU			Sneakers	Adidas					
14	08-DEC-DE	08-DEC-DE			Coffee Maker	Nespresso					
15	18-FEB-CA	18-FEB-CA			Smartwatch	Fitbit					
16	16-APR-ES	16-APR-ES			Headphones	Bose					
17	21-AUG-CA	21-AUG-CA			Laptop Bag	Samsonite					
18	20-AUG-CN	20-AUG-CN			Smartwatch	Huawei					
19	27-JAN-IT	27-JAN-IT			Laptop	Asus					
20	01-MAR-UK	01-MAR-UK			Sunglasses	Oakley					
21	14-AUG-US	14-AUG-US			Camping Tent	Coleman					
22	14-AUG-US	14-AUG-US			Camera	Coleman					
23	09-JAN-CA	09-JAN-CA			Microwave	Panasonic	80	20	Kitchen	80	Kitchen
24	19-JUL-BR	19-JUL-BR			Fitness Tracker	Xiaomi	150	30	Electronics	150	Electronics
25	21-AUG-CA	21-AUG-CA			Laptop Bag	Samsonite		35	Accessories	290	Accessories

Convert Text to Columns Wizard - Step 1 of 3

The Text Wizard has determined that your data is Delimited.
If this is correct, choose Next, or choose the data type that best describes your data.

Original data type

Choose the file type that best describes your data:

☒ Delimited - Characters such as commas or tabs separate each field.
☐ Fixed width - Fields are aligned in columns with spaces between each field.

Preview of selected data:

2 28-JAN-US
3 15-FEB-US
4 03-MAR-US
5 11-APR-US
6 22-MAY-US

Cancel < Back Next > Finish

Product ID	Product ID	anufacturing Da	Country Code	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)	Category.
28-JAN-US	28-JAN-US			Laptop	Dell	1000	30	Electronics	1000	Electronics
15-FEB-US	15-FEB-US			Sneakers	Nike	80	15	Fashion	80	Fashion
03-MAR-US	03-MAR-US			Coffee Maker	Keurig					
11-APR-US	11-APR-US			Smartphone	Samsung					
22-MAY-US	22-MAY-US			Backpack	North Face					
07-JUN-UK	07-JUN-UK			Headphones	Sony					
19-JUL-UK	19-JUL-UK			T-shirt	Adidas					
23-AUG-UK	23-AUG-UK			Blender	Ninja					
05-SEP-UK	05-SEP-UK			Tablet	Apple					
14-OCT-UK	14-OCT-UK			Hiking Boots	Timberland					
07-JUN-UK	07-JUN-UK			Laptop	Sony					
25-NOV-AU	25-NOV-AU			Sneakers	Adidas					
08-DEC-DE	08-DEC-DE			Coffee Maker	Nespresso					
18-FEB-CA	18-FEB-CA			Smartwatch	Fitbit					
16-APR-ES	16-APR-ES			Headphones	Bose					
21-AUG-CA	21-AUG-CA			Laptop Bag	Samsonite					
20-AUG-CN	20-AUG-CN			Smartwatch	Huawei					
27-JAN-IT	27-JAN-IT			Laptop	Asus					
01-MAR-UK	01-MAR-UK			Sunglasses	Oakley					
14-AUG-US	14-AUG-US			Camping Tent	Coleman					
14-AUG-US	14-AUG-US			Camera	Coleman					
09-JAN-CA	09-JAN-CA			Microwave	Panasonic	80	20	Kitchenn	80	Kitchenn
19-JUL-BR	19-JUL-BR			Fitness Tracker	Xiaomi	150	30	Electronics	150	Electronics
21-AUG-CA	21-AUG-CA			Laptop Bag	Samsonite		35	Accessories	298	Accessories
29-SEP-CA	29-SEP-CA			Smartphone	Google	800	45		800	Unknown

Convert Text to Columns Wizard - Step 2 of 3

This screen lets you set the delimiters your data contains. You can see how your text is affected in the preview below.

Delimiters

☐ Tab

☐ Semicolon

☒ Comma

☐ Space

☒ Other: -

☐ Treat consecutive delimiters as one

Text qualifier: -

Data preview

28 JAN US
15 FEB US
03 MAR US
11 APR US
22 MAY US

Cancel < Back Next > Finish

Product ID	Product ID	anufacturing Da	Country Code	Product Name	Brand Name	Price (\$)	Quantity	Category	Price (\$)	Category.
28-JAN-US	28-JAN-US			Laptop	Dell	1000	30	Electronics	1000	Electronics
15-FEB-US	15-FEB-US			Sneakers	Nike	80	15	Fashion	80	Fashion
03-MAR-US	03-MAR-US			Coffee Maker	Keurig					
11-APR-US	11-APR-US			Smartphone	Samsung					
22-MAY-US	22-MAY-US			Backpack	North Face					
07-JUN-UK	07-JUN-UK			Headphones	Sony					
19-JUL-UK	19-JUL-UK			T-shirt	Adidas					
23-AUG-UK	23-AUG-UK			Blender	Ninja					
05-SEP-UK	05-SEP-UK			Tablet	Apple					
14-OCT-UK	14-OCT-UK			Hiking Boots	Timberland					
07-JUN-UK	07-JUN-UK			Laptop	Sony					
25-NOV-AU	25-NOV-AU			Sneakers	Adidas					
08-DEC-DE	08-DEC-DE			Coffee Maker	Nespresso					
18-FEB-CA	18-FEB-CA			Smartwatch	Fitbit					
16-APR-ES	16-APR-ES			Headphones	Bose					
21-AUG-CA	21-AUG-CA			Laptop Bag	Samsonite					
20-AUG-CN	20-AUG-CN			Smartwatch	Huawei					
27-JAN-IT	27-JAN-IT			Laptop	Asus					
01-MAR-UK	01-MAR-UK			Sunglasses	Oakley					
14-AUG-US	14-AUG-US			Camping Tent	Coleman					
14-AUG-US	14-AUG-US			Camera	Coleman					
09-JAN-CA	09-JAN-CA			Microwave	Panasonic	80	20	Kitchenn	80	Kitchenn
19-JUL-BR	19-JUL-BR			Fitness Tracker	Xiaomi	150	30	Electronics	150	Electronics
21-AUG-CA	21-AUG-CA			Laptop Bag	Samsonite		35	Accessories	298	Accessories

Convert Text to Columns Wizard - Step 3 of 3

This screen lets you select each column and set the Data Format.

Column data format

☐ General

☒ Text

☐ Date: DMY

☐ Do not import column (skip)

'General' converts numeric values to numbers, date values to dates, and all remaining values to text.

Advanced...

Destination: \$B\$2

Data preview

28 JAN US
15 FEB US
03 MAR US
11 APR US
22 MAY US

Cancel < Back Next > Finish

Result:

Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Price (\$)	Quantity	Category
28-JAN-US	28	US	Laptop	Dell	1000	30	Electronics
15-FEB-US	15	US	Sneakers	Nike	80	15	Fashion
03-MAR-US	3	US	Coffee Maker	Keurig	130	40	
11-APR-US	11	US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	22	US	Backpack	North Face		20	Outdoor
07-JUN-UK	7	UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	19	UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	23	UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	5	UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	14	UK	Hiking Boots	Timberland	130	10	Outdoor
07-JUN-UK	7	UK	Laptop	Sony	200	45	Electronics
25-NOV-AU	25	AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	8	DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	18	CA	Smartwatch	Fitbit		15	Electronics
16-APR-ES	16	ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	21	CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	20	CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	27	IT	Laptop	Asus	980	10	Electronics

Merging Data:

Merging data is necessary when information is spread across multiple sources or tables. Combining datasets allows for a comprehensive view, enabling better analysis and decision making.

I am using CONCATENATE function to merge data from multiple columns data's into a single column

Syntax: =CONCATENATE(text1, text2)

Here, as an example I am merging the "Brand Name" and "Product Name" columns into one column named "Product Brand".

SUM		=CONCATENATE(E2," ",D2)				
A	B	C	D	E	F	G
Product ID	Manufaturing Date	Country Code	Product Name	Brand Name	Product Brand	Price (\$)
28-JAN-US	28	US	Laptop	Dell	=CONCATENATE(E2," ",D2)	
15-FEB-US	15	US	Sneakers	Nike		8
03-MAR-US	3	US	Coffee Maker	Keurig		13
11-APR-US	11	US	Smartphone	Samsung		90
22-MAY-US	22	US	Backpack	North Face		

Result:

F3

5. Number Formatting:

- Formatting the data type of the "Price" column to a currency format.

In order to format the price column, I have followed the below path

Select the price column->Home->Number tab ->Select \$ symbol

	A	B	C	D	E	F	G	H	I
1	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Product Brand	Price (\$)	Quantity	Category
2	28-JAN-US	28 US		Laptop	Dell	Dell Laptop	1000	30	Electronics
3	15-FEB-US	15 US		Sneakers	Nike	Nike Sneakers	80	15	Fashion
4	03-MAR-US	3 US		Coffee Maker	Keurig	Keurig Coffee Maker	130	40	
5	11-APR-US	11 US		Smartphone	Samsung	Samsung Smartphone	900	25	Electronics
6	22-MAY-US	22 US		Backpack	North Face	North Face Backpack		20	Outdoor
7	07-JUN-UK	7 UK		Headphones	Sony	Sony Headphones	200	45	Electronics
8	19-JUL-UK	19 UK		T-shirt	Adidas	Adidas T-shirt	30	5	Fashion
9	23-AUG-UK	23 UK		Blender	Ninja	Ninja Blender	90	35	Kitchen
10	05-SEP-UK	5 UK		Tablet	Apple	Apple Tablet	500	50	Electronics

Result:

	A	B	C	D	E	F	G	H	I	J
1	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Product Brand	Price (\$)	Quantity	Category	Price
2	28-JAN-US	28 US		Laptop	Dell	Dell Laptop	\$ 1,000.00	30	Electronics	
3	15-FEB-US	15 US		Sneakers	Nike	Nike Sneakers	\$ 80.00	15	Fashion	
4	03-MAR-US	3 US		Coffee Maker	Keurig	Keurig Coffee Maker	\$ 130.00	40		
5	11-APR-US	11 US		Smartphone	Samsung	Samsung Smartphone	\$ 900.00	25	Electronics	
6	22-MAY-US	22 US		Backpack	North Face	North Face Backpack		20	Outdoor	
7	07-JUN-UK	7 UK		Headphones	Sony	Sony Headphones	\$ 200.00	45	Electronics	
8	19-JUL-UK	19 UK		T-shirt	Adidas	Adidas T-shirt	\$ 30.00	5	Fashion	
9	23-AUG-UK	23 UK		Blender	Ninja	Ninja Blender	\$ 90.00	35	Kitchen	
10	05-SEP-UK	5 UK		Tablet	Apple	Apple Tablet	\$ 500.00	50	Electronics	

- Formatting date format of Manufacturing date:

Select manufacturing date column->Home tab->Number tab->more accounting formats->select date->select appropriate format

	A	B	C
1	Product ID	Manufacturing Date	Country Code
2	28-JAN-US	28-Jan-1900	US
3	15-FEB-US	15-Jan-1900	US
4	03-MAR-US	3-Jan-1900	US
5	11-APR-US	11-Jan-1900	US
6	22-MAY-US	22-Jan-1900	US
7	07-JUN-UK	7-Jan-1900	UK
8	19-JUL-UK	19-Jan-1900	UK
9	23-AUG-UK	23-Jan-1900	UK
10	05-SEP-UK	5-Jan-1900	UK

6. Conditional Formatting:

It allows to visually highlight, emphasize or apply specific formatting to cells or range of cells based on certain conditions.

It can also used to emphasize high or low values, identify duplicates, visualize data distribution, and more.

- **Applying data bar or color scales conditional formatting in the "Price" column:**

I have used data bar here. Select range of cells that is price column->in Home Tab->

Conditional Formatting-> Data bars->Gradient Fill

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Product Brand	Price (\$)	Quantity	Category	Price (\$).	C
2	28-JAN-US	28-Jan-1900	US	Laptop	Dell	Dell Laptop	\$ 1,000.00	30	Electronics	\$ 1,000.00	Ele
3	15-FEB-US	15-Jan-1900	US	Sneakers	Nike	Nike Sneakers	\$ 80.00	15	Fashion	\$ 80.00	Fas
4	03-MAR-US	3-Jan-1900	US	Coffee Maker	Keurig	Keurig Coffee Maker	\$ 130.00	40		\$ 130.00	Unk
5	11-APR-US	11-Jan-1900	US	Smartphone	Samsung	Samsung Smartphone	\$ 900.00	25	Electronics	\$ 900.00	Ele
6	22-MAY-US	22-Jan-1900	US	Backpack	North Face	North Face Backpack		20	Outdoor	\$ 297.86	Out
7	07-JUN-UK	7-Jan-1900	UK	Headphones	Sony	Sony Headphones	\$ 200.00	45	Electronics	\$ 200.00	Ele
8	19-JUL-UK	19-Jan-1900	UK	T-shirt	Adidas	Adidas T-shirt	\$ 30.00	5	Fashion	\$ 30.00	Fas
9	23-AUG-UK	23-Jan-1900	UK	Blender	Ninja	Ninja Blender	\$ 90.00	35	Kitchen	\$ 90.00	Kitc
10	05-SEP-UK	5-Jan-1900	UK	Tablet	Apple	Apple Tablet	\$ 500.00	50	Electronics	\$ 500.00	Ele
11	14-OCT-UK	14-Jan-1900	UK	Hiking Boots	Timberland	Timberland Hiking Boots	\$ 130.00	10	Outdoor	\$ 130.00	Out
12	07-JUN-UK	7-Jan-1900	UK	Laptop	Sony	Sony Laptop	\$ 200.00	45	Electronics	\$ 200.00	Ele
13	25-NOV-AU	25-Jan-1900	AU	Sneakers	Adidas	Adidas Sneakers	\$ 90.00	40	Fashion	\$ 90.00	Fas
14	08-DEC-DE	8-Jan-1900	DE	Coffee Maker	Nespresso	Nespresso Coffee Maker	\$ 120.00	35	Kitchen	\$ 120.00	Kitc
15	18-FEB-CA	18-Jan-1900	CA	Smartwatch	Fitbit	Fitbit Smartwatch		15	Electronics	\$ 297.86	Ele
16	16-APR-ES	16-Jan-1900	ES	Headphones	Bose	Bose Headphones	\$ 250.00	20	Electronics	\$ 250.00	Ele
17	21-AUG-CA	21-Jan-1900	CA	Laptop Bag	Samsonite	Samsonite Laptop Bag	\$ 50.00	35	Accessories	\$ 50.00	Acc
18	20-AUG-CN	20-Jan-1900	CN	Smartwatch	Huawei	Huawei Smartwatch	\$ 160.00	15	Electronics	\$ 160.00	Ele
19	27-JAN-IT	27-Jan-1900	IT	Laptop	Asus	Asus Laptop	\$ 980.00	10	Electronics	\$ 980.00	Ele
20	01-MAR-UK	1-Jan-1900	UK	Sunglasses	Oakley	Oakley Sunglasses		15	Fashion	\$ 297.86	Fas
21	14-AUG-US	14-Jan-1900	US	Camping Tent	Coleman	Coleman Camping Tent	\$ 200.00	10	Outdoor	\$ 200.00	Out
22	14-AUG-US	14-Jan-1900	US	Camera	Coleman	Coleman Camera	\$ 200.00	10	Outdoor	\$ 200.00	Out
23	09-JAN-CA	9-Jan-1900	CA	Microwave	Panasonic	Panasonic Microwave	\$ 80.00	20	Kitchenn	\$ 80.00	Kitc
24	19-JUL-BR	19-Jan-1900	BR	Fitness Tracker	Xiaomi	Xiaomi Fitness Tracker	\$ 150.00	30	Electronics	\$ 150.00	Ele
25	21-AUG-CA	21-Jan-1900	CA	Laptop Bag	Samsonite	Samsonite Laptop Bag		35	Accessories	\$ 297.86	Acc
26	29-SEP-CA	29-Jan-1900	CA	Smartphone	Google	Google Smartphone	\$ 800.00	45		\$ 800.00	Unk
27	03-JUN-CA	3-Jan-1900	CA	Sunglasses	Ray-Ban	Ray-Ban Sunglasses	\$ 130.00	25	Fashion	\$ 130.00	Fas
28	11-JUL-CA	11-Jan-1900	CA	Blender	Vitamix	Vitamix Blender	\$ 400.00	40	Kitchen	\$ 400.00	Kitc

Result:

	A	B	C	D	E	F	G	H	I	J	K
1	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Product Brand	Price (\$)	Quantity	Category	Price (\$).	
2	28-JAN-US	28-Jan-1900	US	Laptop	Dell	Dell Laptop	\$ 1,000.00	30	Electronics	\$ 1,000.00	E
3	15-FEB-US	15-Jan-1900	US	Sneakers	Nike	Nike Sneakers	\$ 80.00	15	Fashion	\$ 80.00	F
4	03-MAR-US	3-Jan-1900	US	Coffee Maker	Keurig	Keurig Coffee Maker	\$ 130.00	40		\$ 130.00	U
5	11-APR-US	11-Jan-1900	US	Smartphone	Samsung	Samsung Smartphone	\$ 900.00	25	Electronics	\$ 900.00	E
6	22-MAY-US	22-Jan-1900	US	Backpack	North Face	North Face Backpack		20	Outdoor	\$ 297.86	O
7	07-JUN-UK	7-Jan-1900	UK	Headphones	Sony	Sony Headphones	\$ 200.00	45	Electronics	\$ 200.00	E
8	19-JUL-UK	19-Jan-1900	UK	T-shirt	Adidas	Adidas T-shirt	\$ 30.00	5	Fashion	\$ 30.00	F
9	23-AUG-UK	23-Jan-1900	UK	Blender	Ninja	Ninja Blender	\$ 90.00	35	Kitchen	\$ 90.00	K
10	05-SEP-UK	5-Jan-1900	UK	Tablet	Apple	Apple Tablet	\$ 500.00	50	Electronics	\$ 500.00	E
11	14-OCT-UK	14-Jan-1900	UK	Hiking Boots	Timberland	Timberland Hiking Boots	\$ 130.00	10	Outdoor	\$ 130.00	O
12	07-JUN-UK	7-Jan-1900	UK	Laptop	Sony	Sony Laptop	\$ 200.00	45	Electronics	\$ 200.00	E
13	25-NOV-AU	25-Jan-1900	AU	Sneakers	Adidas	Adidas Sneakers	\$ 90.00	40	Fashion	\$ 90.00	F
14	08-DEC-DE	8-Jan-1900	DE	Coffee Maker	Nespresso	Nespresso Coffee Maker	\$ 120.00	35	Kitchen	\$ 120.00	K

- Creating a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."

In order to do the above task, I have used the conditional formatting as below.

Select range of cells that is Category column->in Home Tab-> Conditional Formatting -> Highlight cell Rules-> More Rule -> Format only cells that contain -> Cell value equal to Electronics -> Format -> Fill -> Choose Color -> Click OK

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Product Brand	Price (\$)	Quantity	Category	Price (\$)	Category
2	28-JAN-US	28-Jan-1900	US	Laptop	Dell	Dell Laptop	\$ 1,000.00	30	Electronics	\$ 1,000.00	Electronics
3	15-FEB-US	15-Jan-1900	US	Sneakers	Nike	Nike Sneakers	\$ 80.00	15	Fashion	\$ 80.00	Fashion
4	03-MAR-US	3-Jan-1900	US	Coffee Maker	Keurig	Keurig Coffee Maker	\$ 130.00	40		\$ 130.00	Unknown
5	11-APR-US	11-Jan-1900	US	Smartphone	Samsung	Samsung Smartphone	\$ 900.00	25	Electronics	\$ 900.00	Electronics
6	22-MAY-US	22-Jan-1900	US	Backpack	North Face	North Face Backpack		20	Outdoor	\$ 297.86	Outdoor
7	07-JUN-UK	7-Jan-1900	UK	Headphones	Sony	Sony Headphones	\$ 200.00	45	Electronics	\$ 200.00	Electronics
8	19-JUL-UK	19-Jan-1900	UK	T-shirt	Adidas	Adidas T-shirt	\$ 30.00	5	Fashion	\$ 30.00	Fashion
9	23-AUG-UK	23-Jan-1900	UK	Blender	Ninja	Ninja Blender	\$ 90.00	35	Kitchen	\$ 90.00	Kitchen
10	05-SEP-UK	5-Jan-1900	UK	Tablet	Apple	Apple Tablet	\$ 500.00	50	Electronics	\$ 500.00	Electronics
11	14-OCT-UK	14-Jan-1900	UK	Hiking Boots	Timberland	Timberland Hiking Boots	\$ 130.00	10	Outdoor	\$ 130.00	Outdoor
12	07-JUN-UK	7-Jan-1900	UK	Laptop	Sony	Sony Laptop	\$ 200.00	45	Electronics	\$ 200.00	Electronics
13	25-NOV-AU	25-Jan-1900	AU	Sneakers	Adidas	Adidas Sneakers	\$ 90.00	40	Fashion	\$ 90.00	Fashion
14	08-DEC-DE	8-Jan-1900	DE	Coffee Maker	Nespresso	Nespresso Coffee Maker	\$ 120.00	35	Kitchen	\$ 120.00	Kitchen
15	18-FEB-CA	18-Jan-1900	CA	Smartwatch	Fitbit	Fitbit Smartwatch		15	Electronics	\$ 297.86	Electronics
16	16-APR-ES	16-Jan-1900	ES	Headphones	Bose	Bose Headphones	\$ 250.00	20	Electronics	\$ 250.00	Electronics
17	21-AUG-CA	21-Jan-1900	CA	Laptop Bag	Samsonite	Samsonite Laptop Bag	\$ 50.00	35	Accessories	\$ 50.00	Accessories

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Product Brand	Price (\$)	Quantity	Category	Price (\$)	Category
2	28-JAN-US	28-Jan-1900	US	Laptop	Dell	Dell Laptop	\$ 1,000.00	30	Electronics	\$ 1,000.00	Electronics
3	15-FEB-US	15-Jan-1900	US	Sneakers	Nike	Nike Sneakers	\$ 80.00	15	Fashion	\$ 80.00	Fashion
4	03-MAR-US	3-Jan-1900	US	Coffee Maker	Keurig	Keurig Coffee Maker	\$ 130.00	40		\$ 130.00	Unknown
5	11-APR-US	11-Jan-1900	US	Smartphone	Samsung	Samsung Smartphone	\$ 900.00	25	Electronics	\$ 900.00	Electronics
6	22-MAY-US	22-Jan-1900	US	Backpack	North Face	North Face Backpack		20	Outdoor	\$ 297.86	Outdoor
7	07-JUN-UK	7-Jan-1900	UK	Headphones	Sony	Sony Headphones	\$ 200.00	45	Electronics	\$ 200.00	Electronics
8	19-JUL-UK	19-Jan-1900	UK	T-shirt	Adidas	Adidas T-shirt	\$ 30.00	5	Fashion	\$ 30.00	Fashion
9	23-AUG-UK	23-Jan-1900	UK	Blender	Ninja	Ninja Blender	\$ 90.00	35	Kitchen	\$ 90.00	Kitchen
10	05-SEP-UK	5-Jan-1900	UK	Tablet	Apple	Apple Tablet	\$ 500.00	50	Electronics	\$ 500.00	Electronics
11	14-OCT-UK	14-Jan-1900	UK	Hiking Boots	Timberland	Timberland Hiking Boots	\$ 130.00	10	Outdoor	\$ 130.00	Outdoor
12	07-JUN-UK	7-Jan-1900	UK	Laptop	Sony	Sony Laptop	\$ 200.00	45	Electronics	\$ 200.00	Electronics
13	25-NOV-AU	25-Jan-1900	AU	Sneakers	Adidas	Adidas Sneakers	\$ 90.00	40	Fashion	\$ 90.00	Fashion
14	08-DEC-DE	8-Jan-1900	DE	Coffee Maker	Nespresso	Nespresso Coffee Maker	\$ 120.00	35	Kitchen	\$ 120.00	Kitchen
15	18-FEB-CA	18-Jan-1900	CA	Smartwatch	Fitbit	Fitbit Smartwatch		15	Electronics	\$ 297.86	Electronics
16	16-APR-ES	16-Jan-1900	ES	Headphones	Bose	Bose Headphones	\$ 250.00	20	Electronics	\$ 250.00	Electronics
17	21-AUG-CA	21-Jan-1900	CA	Laptop Bag	Samsonite	Samsonite Laptop Bag	\$ 50.00	35	Accessories	\$ 50.00	Accessories
18	20-AUG-CN	20-Jan-1900	CN	Smartwatch	Huawei	Huawei Smartwatch	\$ 160.00	15	Electronics	\$ 160.00	Electronics
19	27-JAN-IT	27-Jan-1900	IT	Laptop	Asus	Asus Laptop	\$ 980.00	10	Electronics	\$ 980.00	Electronics

Result:

	A	B	C	D	E	F	G	H	I	J	K
	Product ID	Manufacturing Date	Country Code	Product Name	Brand Name	Product Brand	Price (\$)	Quantity	Category	Price (\$)	Category
2	28-JAN-US	28-Jan-1900	US	Laptop	Dell	Dell Laptop	\$ 1,000.00	30	Electronics	\$ 1,000.00	Electronics
3	15-FEB-US	15-Jan-1900	US	Sneakers	Nike	Nike Sneakers	\$ 80.00	15	Fashion	\$ 80.00	Fashion
4	03-MAR-US	3-Jan-1900	US	Coffee Maker	Keurig	Keurig Coffee Maker	\$ 130.00	40		\$ 130.00	Unknown
5	11-APR-US	11-Jan-1900	US	Smartphone	Samsung	Samsung Smartphone	\$ 900.00	25	Electronics	\$ 900.00	Electronics
6	22-MAY-US	22-Jan-1900	US	Backpack	North Face	North Face Backpack		20	Outdoor	\$ 297.86	Outdoor
7	07-JUN-UK	7-Jan-1900	UK	Headphones	Sony	Sony Headphones	\$ 200.00	45	Electronics	\$ 200.00	Electronics
8	19-JUL-UK	19-Jan-1900	UK	T-shirt	Adidas	Adidas T-shirt	\$ 30.00	5	Fashion	\$ 30.00	Fashion
9	23-AUG-UK	23-Jan-1900	UK	Blender	Ninja	Ninja Blender	\$ 90.00	35	Kitchen	\$ 90.00	Kitchen
10	05-SEP-UK	5-Jan-1900	UK	Tablet	Apple	Apple Tablet	\$ 500.00	50	Electronics	\$ 500.00	Electronics
11	14-OCT-UK	14-Jan-1900	UK	Hiking Boots	Timberland	Timberland Hiking Boots	\$ 130.00	10	Outdoor	\$ 130.00	Outdoor
12	07-JUN-UK	7-Jan-1900	UK	Laptop	Sony	Sony Laptop	\$ 200.00	45	Electronics	\$ 200.00	Electronics
13	25-NOV-AU	25-Jan-1900	AU	Sneakers	Adidas	Adidas Sneakers	\$ 90.00	40	Fashion	\$ 90.00	Fashion
14	08-DEC-DE	8-Jan-1900	DE	Coffee Maker	Nespresso	Nespresso Coffee Maker	\$ 120.00	35	Kitchen	\$ 120.00	Kitchen
15	18-FEB-CA	18-Jan-1900	CA	Smartwatch	Fitbit	Fitbit Smartwatch		15	Electronics	\$ 297.86	Electronics
16	16-APR-ES	16-Jan-1900	ES	Headphones	Bose	Bose Headphones	\$ 250.00	20	Electronics	\$ 250.00	Electronics
17	21-AUG-CA	21-Jan-1900	CA	Laptop Bag	Samsonite	Samsonite Laptop Bag	\$ 50.00	35	Accessories	\$ 50.00	Accessories
18	20-AUG-CN	20-Jan-1900	CN	Smartwatch	Huawei	Huawei Smartwatch	\$ 160.00	15	Electronics	\$ 160.00	Electronics
19	27-JAN-IT	27-Jan-1900	IT	Laptop	Asus	Asus Laptop	\$ 980.00	10	Electronics	\$ 980.00	Electronics